Square-root remainders with logarithmic factors for generalized quantum equipartition

Abstract

Under compactness, convexity, permutation symmetry, primal and positive-polar tensor closure, and uniform growth assumptions, we prove a finite-error quantum equipartition estimate with remainder $O_\varepsilon(\sqrt{n}\,[\log(n+1)]^{3/2})$. The estimate uses the regularized Umegaki rate and is accompanied by an exponentially weighted likelihood-moment bound for every exact entropy-minimizing pair.

We prove a finite-error equipartition estimate for two families of matrices on tensor powers of a fixed finite-dimensional Hilbert space. The testing quantity is the infimum of the usual pairwise hypothesis-testing divergence over both families. The same regularized Umegaki rate controls both sides of the estimate and the exponentially weighted likelihood moment of every exact joint entropy minimizer.

Fang, Fawzi, and Fawzi established generalized equipartition with a finite-error remainder of order $n^{2/3}\log n$ [1]. Under the assumptions displayed below, our remainder is $O_\varepsilon(\sqrt{n}\,[\log(n+1)]^{3/2})$. Thus the improvement is in the power of n, with logarithmic factors remaining; this is neither a purely logarithmic remainder nor an $O(\sqrt{n})$ estimate.

For a fixed i.i.d. pair, Tomamichel–Hayashi [3] and Li [4] give the second-order expansion

$$D_{\mathrm{H}}^{\varepsilon}(\rho^{\otimes n}\|\sigma^{\otimes n}) = nD(\rho\|\sigma) + \sqrt{nV(\rho\|\sigma)}\,\Phi^{-1}(\varepsilon) + O(\log n),$$

where V is the relative-entropy variance and Φ the standard normal distribution function. For full-rank pairs with $V > 0$ and $\varepsilon \neq 1/2$, these singleton families already require a remainder of order $\sqrt{n}$. The present estimate approaches that scale but retains logarithmic factors. Generalized Stein lemmas instead identify first-order rates against composite alternative families; see Lami [5]. The result below concerns a quantitative remainder under the stated primal and polar tensor-closure assumptions.

1 Families, divergences, and the theorem

Let $H = \mathbb{C}^d$, with $d \geq 1$. For each integer $n \geq 1$, let $\mathcal{A}_n$ be a nonempty compact convex set of density matrices on $H^{\otimes n}$, and let $\mathcal{B}_n$ be a nonempty compact convex set of positive semidefinite matrices on that space. Both families are invariant under permutations of the tensor factors. For a set S of positive matrices, define its positive polar by

$$S^{\circ,+} = \{X \geq 0 : \operatorname{Tr}(YX) \leq 1 \text{ for every } Y \in S\}.$$

Assume, for all $m, n \geq 1$, the four inclusions

$$\mathcal{A}_m \otimes \mathcal{A}_n \subseteq \mathcal{A}_{m+n}, \qquad \mathcal{B}_m \otimes \mathcal{B}_n \subseteq \mathcal{B}_{m+n}, \tag{1.1}$$

$$\mathcal{A}_m^{\circ,+} \otimes \mathcal{A}_n^{\circ,+} \subseteq \mathcal{A}_{m+n}^{\circ,+}, \qquad \mathcal{B}_m^{\circ,+} \otimes \mathcal{B}_n^{\circ,+} \subseteq \mathcal{B}_{m+n}^{\circ,+}. \tag{1.2}$$

Finally, assume there is a finite constant C such that every $n \geq 1$ and every $\rho \in \mathcal{A}_n$, $\sigma \in \mathcal{B}_n$ satisfy

$$\rho \leq 2^{Cn}\sigma, \qquad \log_2 \operatorname{Tr}\sigma \leq Cn. \tag{1.3}$$

We may enlarge C so that $C \geq 1$. In particular $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$. Alternatives need not be normalized or invertible.

All divergences below use base-two logarithms. For a supported pair define

$$D(\rho\|\sigma) = \operatorname{Tr} \rho(\log_2 \rho - \log_2 \sigma).$$

The logarithms are evaluated on their positive spectral supports; zero-weight terms are zero. Set $D = +\infty$ when support inclusion fails. Write

$$\begin{aligned} D_n &= \inf_{\rho\in\mathcal{A}_n,\,\sigma\in\mathcal{B}_n} D(\rho\|\sigma), \\ \beta_\varepsilon(\rho\|\sigma) &= \min\{\operatorname{Tr}\sigma Q : 0 \le Q \le I,\ \operatorname{Tr}\rho Q \ge 1-\varepsilon\}, \\ D_{\mathrm{H}}^{\varepsilon}(\rho\|\sigma) &= -\log_2 \beta_\varepsilon(\rho\|\sigma), \\ H_n^{\varepsilon} &= \inf_{\rho\in\mathcal{A}_n,\,\sigma\in\mathcal{B}_n} D_{\mathrm{H}}^{\varepsilon}(\rho\|\sigma). \end{aligned}$$

Thus the minimizing test may depend on the pair. Define

$$r = \inf_{n\ge 1} \frac{D_n}{n}, \qquad \ell_n = \log_2(n+1). \tag{1.4}$$

For spectral decompositions $\rho = \sum_i p_i |u_i\rangle\langle u_i|$ and $\sigma = \sum_j q_j |v_j\rangle\langle v_j|$, use the Nussbaum–Szkoła law [2]

$$w_{ij} = p_i |\langle u_i, v_j\rangle|^2, \qquad Z_{ij} = \log_2 p_i - \log_2 q_j. \tag{1.5}$$

It is a probability law. If $p_i = 0$ or $q_j = 0$, its relevant weight is zero: in the second case this follows from support inclusion. Assign any finite value to Z on those zero-weight atoms. Expectations in the theorem refer to this law, for the particular pair in question.

Theorem 1.1. *Under (1.1)–(1.3), the following statements hold.*

1. *Each D_n is finite, its infimum is attained, and the entire sequence D_n/n converges to the finite number r in (1.4).*
2. *For every fixed $\varepsilon \in (0,1)$ there is a finite $K_\varepsilon > 0$ such that, for every $n \ge 1$, H_n^ε is finite and*

$$\left|H_n^\varepsilon - nr\right| \le K_\varepsilon \sqrt{n}\, \ell_n^{3/2}. \tag{1.6}$$

3. *There are finite $K, c > 0$ such that every $n \ge 1$, every exact minimizing pair $D(\rho\|\sigma) = D_n$, and every $0 \le t \le c/n$ satisfy*

$$\mathbb{E}\big[(Z-nr)^2 e^{t|Z-nr|}\big] \le K n \ell_n^3. \tag{1.7}$$

The constants K, c are chosen before the length, the minimizing pair, and t.

2 Quantum entropy inputs and pair repetition

For clarity, we record the quantum entropy inputs used in the proof. For $\alpha \ne 1$ put

$$\begin{aligned} D_{\mathrm{P},\alpha}(\rho\|\sigma) &= \frac{1}{\alpha-1}\log_2 \operatorname{Tr}\rho^\alpha\sigma^{1-\alpha}, \\ D_{\mathrm{S},\alpha}(\rho\|\sigma) &= \frac{1}{\alpha-1}\log_2 \operatorname{Tr}\left(\sigma^{(1-\alpha)/(2\alpha)}\rho\sigma^{(1-\alpha)/(2\alpha)}\right)^\alpha. \end{aligned}$$

Negative powers are evaluated on the support of σ, and above-one divergences are $+\infty$ for unsupported pairs. Let $D_{\mathrm{M},\alpha}$ denote the supremum of the classical Rényi divergence over finite POVMs. At order one these are Umegaki and measured relative entropy. For each divergence, its set version is the infimum over both matrix arguments. Write $M_n(\alpha)$ and $S_n(\alpha)$ for the measured and sandwiched set versions.

The tensor, pinching, and finite-estimation results of Fang–Fawzi–Fawzi [1, Lemmas 21, 23, 26, 28, 29] give, for $1/2 \le \alpha < 1$,

$$M_{m+n}(\alpha) \ge M_m(\alpha) + M_n(\alpha), \tag{2.1}$$

$$M_n(\alpha) \ge S_n(\alpha) - g(n), \tag{2.2}$$

$$nr \le D_n \le nr + g(n), \qquad g(n) = 2(d^2+d)\log_2(n+d). \tag{2.3}$$

Umegaki set entropy is subadditive. These results have exactly the compactness, convexity, permutation, primal, and positive-polar hypotheses stated above; their positive alternatives are allowed to be unnormalized. Choose a fixed G such that $g(n) \le G\ell_n$ for all $n \ge 1$.

The state comparisons used below are

$$D_{\mathrm{M},\alpha} \le D_{\mathrm{P},\alpha} \le D \quad (1/2 \le \alpha < 1), \tag{2.4}$$

$$D_{\mathrm{S},1+\delta} \le D_{\mathrm{P},1+\delta}, \tag{2.5}$$

$$D_{\mathrm{S},1-\delta}(\rho\|\sigma) \ge (1-\delta)D_{\mathrm{P},1-\delta}(\rho\|\sigma) - \delta\log_2 \operatorname{Tr}\sigma. \tag{2.6}$$

The last trace term must be retained. The comparisons and the one-shot inequality used later appear in [1, Lemma 12 and Eq. (269)] and follow from the usual Petz and sandwiched trace comparisons on supports.

For completeness, finiteness and the rate limit can already be settled. Write $\tau = \sigma/\operatorname{Tr}\sigma$. Nonnegativity of $D(\rho\|\tau)$ and (1.3) give $D(\rho\|\sigma) \ge -Cn$, while monotonicity of the logarithm gives $D(\rho\|\sigma) \le Cn$. Compactness and lower semicontinuity give an exact minimizer. Tensor additivity of D and (1.1) imply $D_{m+n} \le D_m + D_n$. Fekete's lemma therefore yields the full-sequence limit $D_n/n \to r$, with $|r| \le C$. The upper estimate in (2.3) follows by combining order-one measured superadditivity with the order-one pinching estimate. These observations establish part 1 of Theorem 1.1.

For $0 < \delta \le 1/2$ define the actual measured rate

$$R(\delta) = \lim_{k\to\infty} \frac{M_k(1-\delta)}{k} = \sup_{m\ge 1} \frac{M_m(1-\delta)}{m}. \tag{2.7}$$

The limit exists by (2.1). A one-outcome measurement bounds the sequence below by $-Ck$, and (2.4) bounds it above by $D_k \le Ck$. In particular $R(\delta) \le r$.

Lemma 2.1 (Repetition transfer)**.** *Every admissible pair at length n obeys*

$$D_{\mathrm{P},1-\delta}(\rho\|\sigma) \ge nR(\delta). \tag{2.8}$$

Proof. Primal closure admits $(\rho^{\otimes k}, \sigma^{\otimes k})$ at length kn. Thus

$$M_{kn}(1-\delta) \le D_{\mathrm{M},1-\delta}(\rho^{\otimes k}\|\sigma^{\otimes k}) \le kD_{\mathrm{P},1-\delta}(\rho\|\sigma).$$

Divide by kn and take the full rate limit along the multiples of n. The conclusion applies to every original pair; no tensor factorization of a minimizer is needed. □

3 Likelihood curvature and a fixed finite baseline

For $u > -1$, the spectral identities and half-order bounds of (1.5) are

$$\mathbb{E}Z = D(\rho\|\sigma), \qquad \mathbb{E}2^{uZ} = \operatorname{Tr}\rho^{1+u}\sigma^{-u}, \qquad \mathbb{E}2^{\pm Z/2} \le 2^{Cn/2}. \tag{3.1}$$

The positive half-order bound follows from $\sigma \ge 2^{-Cn}\rho$ and inverse-square-root antitonicity. One may apply it to strictly positive regularizations preserving this domination and pass to the supported limit. The negative half-order bound is $\operatorname{Tr}\sqrt{\rho}\sqrt{\sigma} \le \sqrt{\operatorname{Tr}\rho\ \operatorname{Tr}\sigma}$. Hence, for

$$X = Z - nr, \qquad s = \mathbb{E}X = D(\rho\|\sigma) - nr \ge 0,$$

Markov's inequality gives

$$\mathbb{P}\{X > x\} \le 2^{Cn - x/2}, \qquad \mathbb{P}\{-X > x\} \le 2^{Cn-x/2} \quad (x \ge 0). \tag{3.2}$$

All bounds are uniform over admissible pairs, including singular pairs.

We use the exact remainder

$$R_0(y) = e^{-y} - 1 + y = y^2 \int_0^1 (1-v)e^{-vy}\,dv \ge 0. \tag{3.3}$$

For $J = 4(C+1)n$ and $t \le a/n$, with fixed sufficiently small $a > 0$, integrating (3.2) outside $[-J, J]$ and bounding the integrand inside that interval yields

$$\mathbb{E}[X^2 e^{t|X|}] \le K_0 n^2. \tag{3.4}$$

Indeed, the inner contribution is at most $J^2 e^{aJ/n}$. The two outer contributions are bounded by the boundary value plus the integrals of $(2x + tx^2)e^{tx}2^{Cn-x/2}$, each of which is $O(n^2 2^{-n})$ after shrinking a. Taylor's estimate

$$e^{\pm tX} \le 1 \pm tX + \tfrac{1}{2}t^2X^2e^{t|X|} \tag{3.5}$$

then gives a below-one Petz loss $O(\delta n^2)$ uniformly over all pairs, where $t = \delta \ln 2$. For an exact minimizing pair, $s \le g(n)$ and the positive version gives the corresponding above-one upper bound. Applying (2.2)–(2.6) produces finite constants $A, B, c_0, N_0 > 0$ with

$$M_n(1-\delta) \ge nr - A\ell_n - B\delta n^2 \quad (n \ge N_0,\ 0 < \delta \le c_0/n). \tag{3.6}$$

We next improve this baseline a fixed number of times. Suppose for one fixed $q \in (0, 1]$ there are fixed A, B, c, N such that

$$M_m(1-\delta) \ge mr - A\ell_m - B\delta m^{1+q}\ell_m^{1-q} \quad (m \ge N,\ 0 < \delta \le c/m). \tag{3.7}$$

Put $\gamma = 1/(1+q) \in [1/2, 1)$. From (2.7),

$$r - R(\delta) \le A\ell_m/m + B\delta m^q \ell_m^{1-q}.$$

Choose $m = \lceil \delta^{-\gamma} L^{1-\gamma} \rceil$, where $L = \log_2(1/\delta)$. As $\delta \downarrow 0$, the block window holds because $\delta m \to 0$, and ℓ_m is comparable to L. The two terms have the same order, so

$$0 \le r - R(\delta) \le K\delta^\gamma L^\gamma. \tag{3.8}$$

Fix $t_0 = \zeta/n$, with sufficiently small constant $\zeta > 0$, and put $\delta_n = t_0/\ln 2$. Lemma 2.1 and (3.8) give

$$\ln \mathbb{E}e^{-t_0 X} = t_0\{nr - D_{\mathrm{P},1-\delta_n}(\rho\|\sigma)\} \le K_1 n^{-\gamma}\ell_n^\gamma =: a_n. \tag{3.9}$$

For large n, $e^{a_n}-1\le 2a_n$. On the negative half-line, (3.3) gives $R_0(-u)\ge u^2e^{u/2}/16$ by restricting the integral to $[1/2,3/4]$. On $0\le y\le t_0J$, it gives $R_0(y)\ge e^{-t_0J}y^2/2$. Consequently, for $0\le t\le t_0/2$,

$$\mathbb{E}[X^2e^{t|X|};X\le J]\le k_1t_0^{-2}\mathbb{E}R_0(t_0X)\le k_1t_0^{-2}(2a_n+t_0s). \tag{3.10}$$

The positive extreme tail beyond J is $O(n^22^{-n})$ by (3.2). Thus

$$\mathbb{E}[X^2e^{t|X|}]\le K_2\{ns+n^{2-\gamma}\ell_n^\gamma\}\quad(0\le t\le c_1/n). \tag{3.11}$$

Retaining s is essential. In the negative Taylor estimate its coefficient is $-t+(K_2/2)t^2n$, which is nonpositive for $t\le 1/(K_2n)$. Dropping that term, applying $\ln(1+z)\le z$, and then (2.2) and (2.6), proves (3.7) with

$$q'=1-\gamma=\frac{q}{1+q}.$$

The scale term costs at most $2Cn\delta$ and is absorbed because $n^{2-\gamma}\ell_n^\gamma\ge n$. Starting at (3.6), three applications give $q=1/4$. We therefore fix, once and for all, constants $G,B,c_*,N_*>0$ with $G\ell_n\ge g(n)$ and

$$M_n(1-\delta)\ge nr-G\ell_n-B\delta n^{5/4}\ell_n^{3/4}\quad(n\ge N_*,\ 0<\delta\le c_*/n). \tag{3.12}$$

These constants are fixed before the coefficient chosen in the next section.

4 A single-coefficient logarithmic induction

We prove that there are one finite $M>0$ and one fixed $c>0$ such that

$$M_n(1-\delta)\ge nr-G\ell_n-M\delta n\ell_n^3\quad(n\ge N_*,\ 0<\delta\le c/(n\ell_n)). \tag{4.1}$$

Choose a positive ζ satisfying

$$\zeta\le\min\left\{\frac{\ln 2}{8},\ \frac{1}{100(C+1)},\ 8\ln 2\,c_*\right\},\qquad c=\frac{\zeta}{8\ln 2}. \tag{4.2}$$

At a large current length n set

$$m=\lfloor n/16\rfloor,\qquad t_0=\frac{\zeta}{n\ell_n},\qquad \delta_n=t_0/\ln 2.$$

For all sufficiently large n,

$$n/32\le m<n,\qquad \ell_m\le\ell_n-2,\qquad \delta_nm\ell_m\le\frac{\zeta}{16\ln 2}<c. \tag{4.3}$$

Thus the induction hypothesis at m is applicable at the actual order δ_n.

By (2.7), the hypothesis at m and Lemma 2.1 yield, for every pair at the current length,

$$\ln\mathbb{E}e^{-t_0X}\le t_0n\{G\ell_m/m+M\delta_n\ell_m^3\}=:a_n, \tag{4.4}$$

$$a_n\le\frac{32\zeta G}{n}+\frac{M\zeta^2}{\ln 2}\frac{\ell_n}{n}. \tag{4.5}$$

Only the estimate at the strictly smaller length has been used.

To preserve the leading coefficient, use a sharper scalar remainder bound. In (3.3), the probability density $2(1-v)$ on $[0,1]$ has mean $1/3$. Jensen's inequality therefore gives, for $u \geq 0$,

$$R_0(-u) \geq \tfrac{1}{2}u^2 e^{u/3}, \qquad R_0(u) \geq \tfrac{1}{2}u^2 e^{-u/3}. \tag{4.6}$$

Let $J = 4(C+1)n$ and $T = t_0 J = 4\zeta(C+1)/\ell_n$. If $0 \leq t \leq t_0/3$, these inequalities imply

$$\mathbb{E}[X^2 e^{t|X|}; X \leq J] \leq \frac{2e^{2T/3}}{t_0^2}\mathbb{E}R_0(t_0 X). \tag{4.7}$$

For $X \leq 0$ the factor $2/t_0^2$ is sufficient; for $0 \leq X \leq J$ the remaining factor is at most $e^{2T/3}$. The positive tail still obeys

$$\mathbb{E}[X^2 e^{tX}; X > J] \leq K_f n^2 2^{-n}. \tag{4.8}$$

If $0 \leq a_n \leq 1$, then $e^{a_n} - 1 \leq a_n(1+2a_n)$. Define

$$H_n = e^{2T/3}(1+2a_n), \qquad P = 32G/\zeta.$$

Substitution of (4.4) into (4.7), retaining ℓ_m^3 in its leading term, yields

$$\mathbb{E}[X^2 e^{t|X|}] \leq 2H_n \left\{ \frac{M}{\ln 2} n\ell_m^3 + Pn\ell_n^2 + \frac{n\ell_n}{\zeta}s \right\} + K_f n^2 2^{-n}. \tag{4.9}$$

Here $0 \leq t \leq t_0/3$.

For an order δ in (4.1), $t = \delta \ln 2 \leq t_0/8$. The coefficient of s in the negative Taylor bound is

$$-t + H_n t^2 n\ell_n/\zeta \leq -t(1 - H_n/8).$$

It is nonpositive whenever $H_n \leq 2$. The factor 2 in (4.9) cancels the factor $1/2$ in (3.5). The below-one Petz loss is consequently bounded by

$$\delta \left\{ H_n M n\ell_m^3 + H_n \ln 2\, Pn\ell_n^2 + \tfrac{1}{2}\ln 2\, K_f n^2 2^{-n} \right\}.$$

State comparison adds at most $2Cn\delta$ and set pinching adds $g(n) \leq G\ell_n$. It follows that a fixed K_0, independent of M, n and the pair, closes the current induction step provided

$$H_n M\ell_m^3 + K_0\ell_n^2 \leq M\ell_n^3. \tag{4.10}$$

We now choose M without a circular dependence. Put $N = \lceil M^2 \rceil$. For $N_* \leq n \leq N$, the quarter-stage estimate (3.12) proves (4.1), because $c \leq c_*$ and

$$Bn^{1/4}\ell_n^{3/4} \leq B2^{1/4}M^{1/2}\ell_n^3 \leq M\ell_n^3$$

for sufficiently large $M \geq 1$. This base interval requires only a sublinear power $M^{1/2}$ of the coefficient.

Enlarge M so that $N \geq 32N_*$ and exceeds all fixed thresholds in (4.3). For $n > N$, eventual monotonicity of ℓ_n^2/n and (4.5) show uniformly that

$$a_n\ell_n = O\left(\frac{G\ell_N}{N} + \frac{M\ell_N^2}{N} \right) \longrightarrow 0 \quad \text{as } M \to \infty.$$

Thus we can arrange $a_n \leq 1/(100\ell_n)$ for all $n > N$. The choice (4.2) then implies

$$H_n \leq 1 + \frac{1}{4\ell_n} \leq 2. \tag{4.11}$$

For $\ell_n \geq 8$ and $\ell_m \leq \ell_n - 2$, expansion of the cubic gives

$$\left(1 + \frac{1}{4\ell_n}\right) \ell_m^3 \leq \ell_n^3 - 2\ell_n^2. \tag{4.12}$$

Finally take $M \geq K_0/2$. Equations (4.11) and (4.12) imply (4.10). Increasing M satisfies all these conditions while preserving the finite-base inequality. Strong induction on n, using $N_* \leq m < n$ beyond the base, proves (4.1) with this one fixed coefficient.

5 The global modulus and the wider moment window

For sufficiently small $\delta > 0$, put $L = \log_2(1/\delta)$ and choose

$$m = \left\lfloor \frac{a}{\delta L} \right\rfloor$$

with a sufficiently small fixed $a > 0$. Then $m \geq N_*$, $\ell_m = O(L)$, and $\delta m \ell_m \leq c$. The supremum formula (2.7) and (4.1) give

$$0 \leq r - R(\delta) \leq G\ell_m/m + M\delta\ell_m^3 \leq K\delta \log_2(1/\delta)^3. \tag{5.1}$$

In particular, its order window is now a fixed interval $(0, \delta_0]$ independent of the current length.

Take $t_0 = \zeta'/n$ for a fixed sufficiently small $\zeta' > 0$, and let $\delta_n = t_0/\ln 2$. Repetition transfer and (5.1) imply

$$\ln \mathbb{E} e^{-t_0 X} \leq t_0 n\{r - R(\delta_n)\} \leq K' \frac{\ell_n^3}{n}.$$

The right side tends to zero. Applying the fixed-scale curvature argument (3.10), with $t_0 J$ now constant, and the positive extreme tail gives

$$\mathbb{E}[X^2 e^{t|X|}] \leq K''\{ns + n\ell_n^3\} \quad (0 \leq t \leq c'/n) \tag{5.2}$$

for all large n and every admissible pair. For every exact minimizer, (2.3) gives $0 \leq s \leq g(n) \leq G\ell_n$, so (5.2) proves (1.7) at large lengths. The uniform half-order tails (3.2) bound the same exponential moments over the finitely many remaining lengths, after shrinking c' if necessary and enlarging K. This proves part 3, with a single window c/n and constants uniform over all exact minimizers.

6 Both testing bounds at the same rate

Fix $\varepsilon \in (0, 1)$ and set $f_n = \sqrt{n}\,\ell_n^{3/2}$. The Petz one-shot lower bound [1, Lemma 12] is

$$D_{\mathrm{H}}^{\varepsilon}(\rho\|\sigma) \geq D_{\mathrm{P},1-\delta}(\rho\|\sigma) - \frac{1-\delta}{\delta} \log_2(1/\varepsilon). \tag{6.1}$$

Choose $\delta = 1/f_n$ for sufficiently large n. This belongs to the fixed modulus window, and $\log_2(1/\delta) = O(\ell_n)$. Lemma 2.1 and (5.1) yield, uniformly before taking either family infimum,

$$D_{\mathrm{H}}^{\varepsilon}(\rho\|\sigma) \geq nr - Kn\delta \log_2(1/\delta)^3 - \delta^{-1} \log_2(1/\varepsilon) \geq nr - K_\varepsilon f_n.$$

The same lower bound therefore holds for H_n^{ε}.

For the upper bound we give a physical converse that uses only the second moment in part 3. Let $0 \le Q \le I$, $\operatorname{Tr}\rho Q \ge 1-\varepsilon$, and $\beta = \operatorname{Tr}\sigma Q$. For any $a \in \mathbb{R}$ and any $b > 0$,

$$\mathbb{P}\{Z \le a\} \le (1+b)\varepsilon + (1+b^{-1})2^a\beta. \tag{6.2}$$

To see this, split each overlap as

$$\langle u_i, v_j\rangle = \langle u_i, (I-Q)v_j\rangle + \langle u_i, Qv_j\rangle.$$

Use $|z+w|^2 \le (1+b)|z|^2 + (1+b^{-1})|w|^2$, multiply by p_i, and sum over the event $p_i \le 2^a q_j$. Enlarging each nonnegative sum to all spectral indices gives

$$(1+b)\operatorname{Tr}\rho(I-Q)^2 + (1+b^{-1})2^a \operatorname{Tr}\sigma Q^2.$$

Because $(I-Q)^2 \le I-Q$ and $Q^2 \le Q$, this is at most the right side of (6.2). Kernel columns have weight zero and do not affect the argument.

Choose an exact minimizing pair at length n. Set

$$\eta = (1-\varepsilon)/4, \quad b = (1-\varepsilon)/2, \quad h = \sqrt{K/\eta}\, f_n, \quad a = nr + h,$$

where K is the constant in (1.7). Chebyshev's inequality at $t = 0$ gives $\mathbb{P}\{Z \le a\} \ge 1-\eta$, and $1-\eta-(1+b)\varepsilon > 0$. Thus every accepting effect obeys

$$\beta \ge 2^{-a}\frac{1-\eta-(1+b)\varepsilon}{1+b^{-1}}.$$

Taking the minimum over effects and then the negative logarithm yields

$$D_{\mathrm{H}}^{\varepsilon}(\rho\|\sigma) \le nr + \sqrt{K/\eta}\, f_n - \log_2 \frac{1-\eta-(1+b)\varepsilon}{1+b^{-1}}. \tag{6.3}$$

The original pair-infimum H_n^{ε} is no larger than the value of this admissible pair. This proves its upper bound about the same nr.

Finally (1.3) gives, for every pair and every positive length,

$$2^{-Cn}(1-\varepsilon) \le \beta_{\varepsilon}(\rho\|\sigma) \le \operatorname{Tr}\sigma \le 2^{Cn}.$$

Consequently both pairwise and set testing values are finite. These inequalities absorb the finitely many lengths excluded when choosing $\delta = 1/f_n$. Enlarging K_{ε} proves (1.6) for every $n \ge 1$ and completes Theorem 1.1.

References

[1] K. Fang, H. Fawzi, and O. Fawzi. *Generalized quantum asymptotic equipartition.* arXiv:2411.04035v4, 2026.

[2] M. Nussbaum and A. Szkoła. The Chernoff lower bound for symmetric quantum hypothesis testing. *The Annals of Statistics* **37**(2), 1040–1057, 2009.

[3] M. Tomamichel and M. Hayashi, *A hierarchy of information quantities for finite block length analysis of quantum tasks*, IEEE Trans. Inf. Theory **59**, 7693–7710 (2013). arXiv:1208.1478.

[4] K. Li, *Second-order asymptotics for quantum hypothesis testing*, Ann. Statist. **42**, 171–189 (2014). arXiv:1208.1400.

[5] L. Lami, *A solution of the generalised quantum Stein's lemma*, IEEE Trans. Inf. Theory **71**, 4454–4484 (2025). arXiv:2408.06410.